

Unit 4: Spaces We Live In

NYS content module: Environmental Design and Management (EDM) **Length:** 25 class periods of 40 minutes (days 91 to 115 of the year; the first 25 days of marking period 3) **CTE themes carried:** Sustainability (primary); Problem Solving and Innovation; Health, Safety, and Wellness; Financial and Consumer Literacy; Career and Community Opportunities; Communication and Interpersonal Relationships (the pitch and the gallery walk) **Unit project:** Design a Room (Design Your Own Bedroom for a client, or Tiny House), scored with Rubric 04 - Design a Room.md **Companion files in this folder:** Calendar 04.md, Vocabulary 04.md, Lessons 4.1 to 4.17, Project 04 - Design a Room.md, Rubric 04 - Design a Room.md, Assessment 04.1 - Home Safety and Conservation Quiz.md (given in the last 15 minutes of Lesson 4.5), Assessment 04.2 - Design Vocabulary Check.md (given in the last 12 minutes of Lesson 4.10 Day 2), Assessment 04.3 - Unit 4 Test.md (given in the last 20 minutes of Lesson 4.17). The Color in the Home frosting lab (Lesson 4.8) is scored on the Lab Rubric pattern from Rubric 01 - Lab Rubric.md, printed inside that lesson. Slides and handouts are in 05 Materials/Unit 04/.

Reconciliation note: this plan was written to the scope and sequence, the year calendar, and the assessment slots the unit brief set, then reconciled to all seventeen lesson pages, the project sheet, the rubric, and the three assessment files as they stand. Where a lesson page and this plan disagree, the lesson page wins; the calendar and this plan follow the lessons' handout names, day splits, and check slots.

1. Why this unit

I have been inside more buildings than most teachers, and not as a guest. For several summers I managed a hotel property. That job taught me that a building is not a thing. It is a system of small repairs and energy bills. A running toilet on the third floor is a water bill. A door that does not seal is a heating bill. A breaker that trips every Saturday night is a call I have to make. Nobody teaches that in school, and then people grow up and rent an apartment and do not know which way to turn the shutoff valve.

In college I worked on a zero-energy home project, and that is where I first saw that the design of a house and the cost of running it are the same problem. In my certification program my graded lesson was Color in the Home, where students painted a paper wall with frosting they mixed themselves and then pitched the room to the class. In my classroom I wrote ten interior design packets, from the eight elements to a scale floor plan of your own bedroom to a room built from secondhand and DIY only. This unit takes all of that and puts it in one place for 11 to 14 year olds.

Here is what I believe. A kid who can read a floor plan, fix a running toilet, and cut an electric bill has power at home. Not power over anyone. Power to help. Most of my students live in apartments or shared houses where a parent works two jobs and a repair waits until somebody can pay for it. If my student can find the breaker, plunge the sink, set the

thermostat, and tell the landlord in one clear sentence what is wrong, that family is better off by Friday.

The unit runs in three moves. First the management half: what a home has to do for the people in it, what can hurt you at home, how to do the seven small repairs that come up most, and how to cut the energy and water a home uses. Then the design half: the elements and principles, color mixed by hand with frosting, texture and pattern, and a floor plan drawn to scale. Then the project: a real client with a real need, a scale plan with every path measured, a furniture budget with tax, a model built from a shoebox, and a 60-second pitch to a classmate who plays the client. Every day from the floor plan lesson to the last build day opens with a floor plan sketch, and that stack of sketches is turned in with the project. By the end, a student has drawn eleven rooms, done five repairs, audited a home, and defended a design to a room full of people.

2. Content identified (the concept map, written out)

2.1 What the NYS module lists for this unit

All 14 lines of the Environmental Design and Management module are taught in this unit. None are skipped. EDM 1. b) is quoted from the NYSED 2023 copy, which prints the line cleanly; the June 2018 PDF had fused two phrases (“Plan ways of maintaining” ran into “Identify common safety hazards”), and the crosswalk (Part E.5) records the fix. Two lessons share the line: Lesson 4.2 teaches identifying the hazards and Lesson 4.3 teaches minimizing risk through maintenance and repair.

Line	Text (verbatim)	Taught in
EDM 1. Healthy, Safe, Sustainable Living Spaces a)	Recognize that family, school, work, and community settings are all part of the individual's broader living space	Lessons 4.1 (the seven jobs of a home, and the state's bigger idea that school, a job, and the neighborhood are part of the living space; Four Corners on needs across the lifespan; six family cards matched to six home listings), 4.11 (the client's home and the people who share it)
EDM 1. b)	Identify common safety hazards found in living spaces and describe ways to minimize risks to individuals, families, and household members (wording per the NYSED 2023 copy; the 2018 PDF fused two phrases)	Lessons 4.2 (the second phrase: fifteen hazards written into a described two-floor home, classified by type, with a fix for each), 4.3 (the first phrase: seven small repair jobs, tool ID, five repair stations, and a twelve-month maintenance

		calendar), 4.17 (the test)
EDM 1. c)	Formulate a safety plan for home, work, and community settings	Lessons 4.2 (the family emergency plan: two ways out of every bedroom, one meeting spot, three numbers, and how the grandmother with a cane gets out), 4.3 (the maintenance calendar, which that lesson cites under this line)
EDM 1. d)	Investigate ways to conserve natural resources in family, school, work, and community settings	Lessons 4.4 (the home energy audit; water at the tap and the shower), 4.5 (waste, reuse, and materials), 4.17 (the test)
EDM 2. Environment and Interior Design a)	Design shared environments and living space zones (for home, school, work, and/or community) which show respect for individual privacy needs and appreciation of large group social needs	Lessons 4.10 (zones on a plan: sleep, work, store, move), 4.11 (the six client cards: a grandfather with a walker, a wheelchair user, a shared sibling room, a baby's room, a tiny house couple, an entry that has to work for everyone; needs sorted from wants and the hidden needs found), 4.12 (zones drawn), 4.14 (the model)
EDM 2. b)	Create a floor plan designed to meet the needs of individuals, families, or communities	Lessons 4.10 (read and draw a plan at 1/4 inch = 1 foot), 4.12 (the project plan at 1/2 inch = 1 foot with every path measured), 4.14 (the model built to the plan)
EDM 2. c)	Incorporate the elements and principles of design into a design plan	Lessons 4.6 (elements), 4.7 (principles), 4.8 (color), 4.9 (texture, pattern, mood board), 4.12 (the principles check on the plan), 4.14 (the color scheme and the justification card), 4.15 (the pitch), 4.17 (the test)
EDM 2. d)	Research how	Lessons 4.4 (where the heat

	environmental and green home factors influence the design of functional, safe and aesthetic living spaces	goes: windows, insulation, the thermostat), 4.5 (green materials and secondhand), 4.13 (the secondhand and DIY columns of the catalog), 4.14 (the GREEN label card on the model)
EDM 2. e)	Predict future housing design incorporating technology and demographic data	Lesson 4.16 Step 4, “Where housing is going,” which is where the line is taught and printed: two demographic facts, two technologies, and one written prediction per student. Lesson 4.1 sets it up with aging in place, multigenerational households, and universal design
EDM 3. Consumer Resources and Finance a)	Construct a budget for implementation of a floor plan design	Lessons 4.13 (the Budget Sheet from the 40-item catalog, with tax, under the client’s cap), 4.15 (the budget total and the hardest trade-off in the pitch), 4.17 (the test’s budget item)
EDM 3. b)	Use consumer skills to select household items	Lessons 4.5 (materials and what they cost to own), 4.13 (new, secondhand, or DIY for every piece; what “cheaper” actually costs)
EDM 4. Career Pathways a)	Explain roles and functions of individuals engaged in environment, housing, and interior design careers	Lesson 4.16 (the project chain: thirteen people from the architect’s stamp to the property manager); also on the test
EDM 4. b)	Investigate education and training requirements and opportunities for career paths in environment, housing, and interior design fields	Lesson 4.16 (the three paths: a degree, an apprenticeship or trade, a license after experience)
EDM 4. c)	Assess personal employability skills for careers in environment,	Lesson 4.16 (the “would I like this” self-check, with evidence from Project 04)

	housing, architecture, and interior design and evaluate personal suitability for these careers	
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2.2 How the CTE themes come in

- Sustainability (primary): Topic 4.1 is built on SUS 3. Energy Conservation (energy sources, renewable or not, conservation defined, methods at home, practiced in class) and SUS 1. Resources b) and e) (sustainability as a factor in a decision; stewardship); the home energy audit in Lesson 4.4 and the waste and materials lesson in 4.5 are where the theme is practiced, and the project requires a GREEN label on the model with a reason for this client (SUS 2. f) on Lesson 4.14).
- Problem Solving and Innovation: the project is the state's own design process, PSI 2. Design Process (Proactive) a), run in order from defining the problem (4.11) through the prototype (4.14) to the pitch (4.15); Lesson 4.3 runs the troubleshooting process, PSI 3. a) a. to f., on a running toilet and a tripped breaker.
- Health, Safety, and Wellness: Lesson 4.2 is the safety lesson of the unit (HSW 3. Personal Safety c) and e), HSW 4. g)): causes of injury by room, a practiced emergency plan, and preplanned procedures; Lesson 4.3 applies tool safety and safety glasses at the repair stations (HSW 4. c) and d)); Lesson 4.1 uses HSW 3. d) for the orderly home.
- Financial and Consumer Literacy: Lesson 4.13 is a budget with a cap, a catalog with three prices per item, and sales tax (FCL 4. c), f)): the same five-step decision model from Unit 0 applied to a desk, a dresser, and an armchair.
- Career and Community Opportunities: Lesson 4.16 runs the careers day on CCO 2. Career Clusters a) and CCO 4. Career Plans b) with thirteen career cards, a Project Chain sheet, and a self-check scored against what the student actually did in the project.
- Communication and Interpersonal Relationships: the 30-second frosting pitch (4.8) and the 60-second five-part project pitch to a classmate playing the client (4.15) require a student to say what they designed and why, answer one client question, and give two stars and a step to four other designs (CIR 1. f)).

2.3 Trends, issues, and news

1. Energy prices and heat pumps. Students hear parents talk about the electric bill and the oil bill. New York State is pushing heat pumps and paying part of the cost through NYSERDA programs, and some students' homes have already switched. Lesson 4.4 puts a heat pump next to a furnace and a window unit and asks what each one costs to run.
2. Tiny homes and accessory dwelling units. Students have seen tiny house videos. Long Island towns have been arguing about whether a family can put an apartment over the garage or in the basement. Lesson 4.1 uses both as housing types, and the Tiny House path in the project is the state's own illustrative activity turned into a design problem with a bigger budget and a harder footprint.
3. Multigenerational households on Long Island. A grandparent moving in, or two families sharing a house because of rent, is normal in many schools. Lesson 4.1 counts

it as a housing need across the lifespan (the no-perfect-fit family on the listings sheet is three generations on one income), and Lesson 4.11's Okafor card is a grandfather with a walker moving into a bedroom that was not built for him.

4. Universal design and aging in place. Lever handles, no-step entries, 36-inch paths, and a shower with no lip help a grandparent, a person in a wheelchair, and a parent carrying groceries. Lesson 4.1 names universal design; Lesson 4.12 makes every path on the plan pass a 30-inch or 36-inch rule and draws the 60-inch turning circle; the last do-now sketch in 4.14 puts a turning circle in any room.
5. Smart home devices. Thermostats you set from a phone, plugs that turn off on a schedule, doorbell cameras. Lesson 4.4 asks whether a smart thermostat saves money or just looks like it does; Lesson 4.16 asks what a house full of these devices means for the electrician and the energy auditor.
6. Secondhand furniture apps. Students already know the apps where a desk sells for \$40 and a couch goes free if you can carry it. Lesson 4.13's catalog prices every item three ways (new, secondhand, DIY) at realistic 2026 Long Island figures, and the hook is one desk at \$149, \$40, and \$35 in parts.

2.4 What students this age find interesting

1. Their own bedroom, and the fact that somebody is finally asking what they would change about it.
2. Fixing something that adults usually pay someone to fix, and being able to say "I know how to do that." The \$185 plumber's invoice for an \$8 part is the hook of Lesson 4.3 for that reason.
3. Frosting. Mixing it, painting with it, and defending the color to the class.
4. Building a model with their hands and walking around a room full of them.
5. The electric bill. Every student has heard someone at home complain about it; this is the unit that says what a kilowatt-hour is and which appliance is the problem.
6. A client with a problem to solve, a budget that is real, and a classmate who gets to ask "where do I put my walker at night?"

2.5 Links to other subjects

- Math: scale ($1/4$ inch = 1 foot in Lesson 4.10, doubled to $1/2$ inch = 1 foot for the project plan in 4.12 so a bedroom fills the sheet; the two conversions are the do now of the test day), area (the square feet of a bedroom, the percent of floor left open as a grade 8 stretch), unit conversions (a twin bed at 3 feet by 6 feet 6 inches placed on a plan in squares), and percent (sales tax at 8.625 percent in 4.13; percent of the bill an appliance takes in 4.4).
- Science: energy and heat transfer (where the heat leaves a house, why insulation works, 4.4), electricity (what a watt is, why a breaker trips and what a GFCI does, 4.3 and 4.4), and water (gallons per minute at a faucet and a shower, 4.4).
- Art: the color wheel, tints and shades, warm and cool (4.8), the elements and principles of design that the art room also teaches (4.6, 4.7), and the isometric grid drawing that is the no-materials alternative to the model (4.14).

- ELA: the client brief's one-sentence design problem ("___ needs a room that ___ within ___," 4.11) and the five-part pitch (4.15) are a claim with evidence; the pitch planner is a persuasion outline the ELA teacher can use.
- Social studies: housing and community (4.1): who lives in what kind of housing and why, what zoning is, why a town argues about accessory apartments, and what a multigenerational household looks like on Long Island.
- Technology: a CAD preview (4.10 Day 2, four minutes): I project a free browser floor-plan tool, drag one wall, drop one bed, and say what the software does not know, which is the client. Four minutes, not ten: Day 2 of Lesson 4.10 also carries the 12-minute Design Vocabulary Check, and ten will not fit. If the full ten is wanted, the lesson's teacher notes say it goes at the end of Day 1 in place of Step 4's share, or in the first ten minutes of Lesson 4.11 before the client cards. Nobody is required to use the tool, and the project plan is on graph paper. If the school has a technology or engineering class, the model is a prototype in the same design process the PSI theme describes.
- Health: the health course covers fire safety and carbon monoxide in its own unit; Lesson 4.2 agrees on the words and goes further into the home itself (see Section 11).

3. Enduring understandings and essential questions

Big ideas:

- A home has jobs to do, and the jobs change as the people in it change. A good home is not the biggest or the newest one. It is the one that does its jobs for the people who live there.
- A home is a system. Every part of it costs something to run and something to fix, and small habits change the bill.
- Most home injuries are preventable, and the person who prevents them is usually the one who notices first.
- Good design is not decoration. It is a plan that fits the people who use the space, the way they move through it, and the money they have.
- Color, light, and arrangement change how a person feels in a room, and a designer chooses them on purpose.
- A designer does not design for themselves. Every space has a client, and the client's needs come before the designer's taste.

Essential questions:

1. What does a home have to do for the people who live in it, and who decides when it is doing enough?
2. What in a home can I fix myself, and how do I know when to call someone?
3. Where does the energy and water in my home go, and what would cut the bill?
4. Why does one room feel calm and another feel loud, before anyone says a word?
5. How do you design a room for somebody who is not you?
6. What does it cost to furnish a room, and what do I give up to stay under budget?

4. Knowledge and skills

Knowledge (facts, vocabulary, ideas):

- The seven jobs of a home (shelter, safety, storage, rest, work, play, belonging) and the state's bigger idea that school, a job, and the neighborhood are part of a person's living space.
- Housing needs across the lifespan: a baby, a teen, a grandparent, a wheelchair user; what each one adds to a plain two-bedroom apartment; universal design as designing once so nobody has to change it later.
- Types of housing: apartment, single-family house, condo, townhouse, manufactured or mobile home, shared housing; rent versus own; what a lease, a landlord, and a mortgage are; why neither is "better."
- The common hazards by room: kitchen (burns, grease, knives in the sink), bathroom (wet floor, no grab bar, medicine in reach, a hair dryer near water), stairs (clutter, a loose rug, no rail, dark), bedroom (a space heater by the bed, cords under the rug, a candle, a blocked window), garage (gasoline near the water heater, a running car), outside (a ladder left up, a pool with no gate, ice); the seven hazard types (burn, fall, fire, electrical, poison, drowning, other).
- What a smoke detector and a carbon monoxide detector do, where they go (every level and every sleeping area), and when to test them (monthly), replace batteries (yearly), and replace the unit (ten years); the Poison Control number.
- The three parts of a family emergency plan: two ways out of every bedroom, one meeting spot outside that everyone can reach (never the driveway), and the three numbers to call.
- The seven small repair jobs: change a light bulb (switch off, cool, check the max watts, LED), reset a tripped breaker (find the one in the middle, fully off, then on, and stop if it trips again), unclog a drain without chemicals, stop a running toilet (the chain, the flapper, the float), tighten a loose screw (match the driver to the head), hang something on a wall (stud or anchor, level, slight upward angle), clean a filter (furnace, range hood, dryer lint).
- The tools: Phillips and flathead screwdrivers, pliers, an adjustable wrench, a level, a tape measure, a sink plunger and a toilet plunger (the flange is the difference); the safety rules at a station.
- The six-step troubleshooting process the state names: what is the problem, what should it do, test to find the cause, fix it, check that the fix worked, stop it happening again.
- The maintenance calendar: monthly, seasonal, and yearly tasks and why each one lands where it does.
- Energy at home: where a home's energy goes (heating and cooling about half, water heating about a fifth, the rest lights, appliances, and electronics); what a watt and a kilowatt-hour are; phantom load; LED versus incandescent bulbs; the thermostat setback; insulation and air leaks; what one bulb costs to run for a year and what one change saves; how to read an electric bill (kWh used, supply charge, delivery charge, fixed charge, total).

- Water at home: where it goes (toilets about a quarter, showers and faucets about a fifth each, laundry, leaks); one drip a second is about 5 gallons a day; the running toilet as the largest silent leak; what a low-flow fixture does.
- Waste, reuse, and materials: the four Rs (reduce, reuse, recycle, compost or “rot”) and why reduce comes first; the five green factors of a living space (natural light, materials, indoor air, water, waste); low-VOC paint, fast furniture, secondhand and durable furnishings.
- The eight elements of design: line, shape, form, space, texture, pattern, color, light.
- The six principles of design: balance (symmetrical and asymmetrical) and visual weight, focal point (emphasis), rhythm, proportion and scale, unity (harmony), contrast.
- Color: the wheel, primary, secondary, tertiary, tint, shade, warm, cool, a palette, a scheme (monochromatic, complementary, analogous), the mood a color sets in a room.
- Texture (visual and actual) and pattern (stripe, plaid, floral, geometric, solid): the three-part layering rule (vary the scale, repeat one color, let one pattern lead); what a mood board is for.
- The floor plan: scale (1/4 inch = 1 foot to learn on; 1/2 inch = 1 foot for the project), the eleven symbols (wall, door with its swing arc, window, closet, stairs, sink, toilet, tub, stove, refrigerator, outlet), zones (sleep, work, store, move), traffic flow, and clearance.
- The traffic flow rules: 36 inches for any path a walker or wheelchair uses and for the front entry, 30 inches everywhere else, a 60-inch turning circle for a wheelchair, the door swing clear, nothing in front of a window that must open, one open side of a bed.
- The design process the state names: define the problem, set criteria and constraints, brainstorm, select, prototype, test, optimize, evaluate; a client, a brief, a need (the design fails without it) versus a want (the design is better with it), and hidden needs.
- A furnishings budget: the client’s cap (\$400 for a room refresh, \$1,500 for the tiny house interior), a catalog with three prices per item (new, secondhand, DIY), sales tax at the county rate, a running total, and the trade-off box.
- The project chain: architect, interior designer, contractor, electrician, plumber, HVAC technician, home inspector, energy auditor, sustainability consultant, landscape designer, furniture maker, real estate agent, property manager; the three paths into them; what each pays, to be refreshed before teaching.

Skills (what students will be able to do):

- List the seven jobs of a home with an example of each; name three needs a baby, a teen, a grandparent, or a wheelchair user adds to a plain apartment.
- Match six family cards to six home listings with a sentence per match that names a need, not a preference, and name what is still missing for the family with no perfect fit.
- Read a described home and find fifteen hazards, classify each by type, and write the fix; say which is most dangerous for a toddler and which for a grandmother with a cane.
- Draw a family emergency plan on the described floor plan with two exits per bedroom, a meeting spot that works for a cane, and the three numbers.

- Name seven tools from the tray and match each to a job; at five stations, drive and remove a screw in a pilot hole with safety glasses on, seat a plunger and feel the seal, measure five lines to the nearest eighth of an inch, level a bracket and hang a frame, find the tripped breaker in the photo and say the reset in order, and change a bulb after reading the max watts; pass the test on every station card.
- Say the six troubleshooting steps in order and run a running toilet through them.
- Place fifteen maintenance tasks on a twelve-month grid and justify three placements by cause and season.
- Compute the yearly cost of one bulb, the savings from switching to LED, and the water lost to one dripping faucet; audit one zone of the classroom and a described apartment on the checklist (lights, phantom loads, heating and cooling, water, appliances), scoring each line OK, free fix, or money fix; rank the ten cheapest fixes by yearly savings, compute payback for the ones that cost money, and justify the top three; find the five numbers on a sample electric bill.
- Sort twelve household items into reduce, reuse, recycle, or compost and defend the calls; for a described room with eight wrong items, name a greener swap for each, the green factor it improves, and whether it is free, cheap, or money, then pick the three swaps that do the most good for the least money.
- Find each of the eight elements in a described living room and in the FACS room itself, say what each one does to the feel of the room, and redraw a rough bedroom so three elements change, with a reason for each.
- Classify eight described room layouts by the principle each shows best and its balance type, with a visual-weight sentence; build a room around an assigned focal point in ten minutes in the Focal Point Game and critique two other teams' rooms on the six-question checklist with a specific fix for each.
- Mix a secondary color, a tint, and a shade from primaries in frosting; plan a palette for an assigned Room Card and mood; paint a wall panel; pitch it in 30 seconds with a family or function reason.
- Classify surfaces as visual or actual texture, judge six texture and pattern combinations against the three-part layering rule, and build a mood board (color, texture, pattern, one object) from real swatches for a drawn room and mood, with a one-sentence justification.
- Read a real floor plan (eleven symbols, scale, dimensions) and trace the traffic path; measure the classroom with a tape to the nearest half foot and draw it to scale; draw a bedroom at $1/4$ inch = 1 foot with the door swing right, cut-outs placed in the four zones, and one path under 30 inches fixed.
- Sketch a floor plan from a prompt in a timed do now, eleven times, from the FACS room to a folded-down net of the client's room to an elevation at $1/2$ inch = 1 foot to a room with a wheelchair turning circle in it.
- Read a client card and identify the client, the room dimensions, the budget, and the one rule that cannot be broken; sort every line into a need or a want; add two hidden needs; write the design problem as "___ needs a room that ___ within ___" and defend it.
- Draw the client's room outline at $1/2$ inch = 1 foot with the door swing, the windows, the closet, the scale, and the client name; place 30 cut-out pieces in zones; shade and

measure every path in inches and circle the ones that fail; draw the turning circle if the client uses a wheelchair; run a peer review checklist on another client's plan (mark, do not fix); revise and write two changes made and one suggestion rejected.

- Price every piece on the plan from the catalog in a new, secondhand, or DIY column, add sales tax, keep a running total at or under the client's cap, and justify three purchase decisions with the five-step model.
- Build a shoebox or foam board model at a stated scale that matches the plan, furnish it from scrap, apply the color scheme, add one texture, and label it GREEN and SAFE with a reason for this client and a justification card for the biggest design choice.
- Deliver the five-part pitch in 60 seconds (the client and the need; the plan and the path; the budget total and the hardest trade-off; the color scheme and the mood; the green feature and the safe feature), answer the client's one question in one sentence, walk the gallery and write four two-stars-and-a-step feedback slips, and self-score on Rubric 04 with one line of evidence per criterion.
- Sort thirteen career cards by path and by workplace, compare three careers on the Project Chain sheet, and rate their own fit for two of them with evidence from the project.

5. Unit objectives

One per topic, upper half of Bloom's:

- Topic 4.1: Students will analyze a home for its seven jobs, its hazards, its energy and water use, and its waste using a family-to-home match, a paper safety hunt with an emergency plan, five repair stations, a maintenance calendar, and a home energy audit, and evaluate which three changes would make that home safer and cheaper to run, with numbers.
- Topic 4.2: Students will evaluate real rooms for the elements and principles of design, create a color palette by mixing frosting to match a room and a mood and justify it in a pitch, and draw a bedroom floor plan to scale with correct symbols and a working traffic path.
- Topic 4.3: Students will design a room for a client by writing the design problem, drawing a scale plan with every path measured against the traffic flow rules, constructing a furnishings budget with tax under the client's cap, building a model that matches the plan, and justifying the design to the client's needs in a five-part 60-second pitch, then evaluate four classmates' designs on feedback slips and their own on the rubric.
- Topic 4.4: Students will compare thirteen careers in the housing project chain by role, path, and pay, and assess their own suitability for two of them with evidence from what they did in the project.

6. Topics, sequence, and timeline

Topic	Days	Lessons	Activity lines (verb first)	Formal check at end of topic
4.1 Safe,	7 (days 91 to	4.1 Housing	list the seven	Assessment

Healthy, Sustainable Homes	97)	Needs Across the Lifespan and Types of Housing (1); 4.2 Home Safety Hunt and Emergency Plans (1); 4.3 Simple Maintenance and Repairs (2); 4.4 Conserving Energy and Water: The Home Energy Audit (2); 4.5 Green Choices: Waste, Reuse, Materials, and the Home Safety and Conservation Quiz (1)	jobs of a home and match the do now to them; move to a corner (baby, teen, grandparent, wheelchair user) and list three needs that person adds; fill the housing types table, rent or own; match six family cards to six home listings with a need in every sentence and name what the no-perfect-fit family still lacks; read the described two-floor home and highlight fifteen hazards, classify each, and write the fix; draw the family emergency plan with two exits per bedroom, a meeting spot that works for a cane, and three numbers; pass the smoke and CO detectors around and hear the test button; fill the seven-jobs	04.1 - Home Safety and Conservation Quiz.md, given in the last 15 minutes of Lesson 4.5 (day 97), covering Lessons 4.1 to 4.5 (the lessons say the seven jobs, the breaker item, the drain item, and a scenario built from the two argued matches are on it)
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			table (job, first step, safety rule, tool); handle and name seven tools; write the six troubleshooting steps and run the running toilet through them; rotate through five repair stations with safety glasses at the screw board; place fifteen maintenance tasks on a twelve-month grid and justify three; fill the two pie charts for where a home's energy and water go and define phantom load and conservation; work the bulb math (watts times hours times days, divided by 1,000, times the price per kWh), the LED savings, and the dripping faucet gallons, then three practice problems in pairs; read a	
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			plug-in meter on a charger with nothing attached and on a game console if the room has one; score a described bedroom on the audit checklist; audit one zone of the classroom with planted wastes, then the described Marchetti-Oyelaran apartment, scoring every line OK, free fix, or money fix; cross out the two fixes over \$50, rank the ten cheapest by yearly savings, compute payback for the five that cost money, and justify the top three; find the five numbers on a sample electric bill; sort twelve household items into reduce, reuse, recycle, or compost; name the five green factors of a living space;	
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			run the Greener Room Swap on a described teen bedroom with eight wrong items (the swap, the green factor, free, cheap, or money) and pick the three swaps that do the most good for the least money; take the quiz	
4.2 Design: Elements, Principles, Color	8 (days 98 to 105)	4.6 Elements of Design in a Real Room (1); 4.7 Principles of Design: Balance, Focal Point, Rhythm, Proportion (2); 4.8 Color in the Home (2, the frosting lab); 4.9 Texture, Pattern, and Mood Boards (1); 4.10 Reading and Drawing a Floor Plan to Scale, and the Design Vocabulary Check (2)	fill the eight-elements table from the slides; run the Elements Hunt in a described living room and in the classroom itself and write what each element does to the feel of the room; redraw a rough bedroom sketch so three elements change and justify each change; send the shoebox note home; fill the six-principles table and name the two kinds of balance; classify eight described	Color in the Home frosting lab scored on the Lab Rubric pattern inside Lesson 4.8 (three objectives plus safety and sanitation, 16 points; cite Rubric 01 - Lab Rubric.md), day 102; Assessment 04.2 - Design Vocabulary Check.md, given in the last 12 minutes of Lesson 4.10 Day 2 (day 105)

			layouts by the principle each shows best and its balance type, with a visual-weight sentence; turn the six principles into the six-question critique checklist; play the Focal Point Game (a room outline, a furniture envelope, a focal point card, ten minutes to build) and critique two other teams' rooms with one specific fix each; label the color wheel with the primaries, the secondaries, and the warm and cool sides; watch the live frosting demo (orange, a peach tint, a brick shade); plan a palette for an assigned Room Card (bathroom calm, kitchen warm, game room energetic, home office focused) or a	
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			Challenge Card; mix at least one secondary, one tint, and one shade in frosting and paint a paper-plate wall panel; pitch it in 30 seconds and vote by thumbs; keep the color card in the project folder for Lesson 4.14; classify six surfaces as visual or actual texture and name the pattern type; judge six texture and pattern combinations against the three-part layering rule; build a mood board from four real swatches for a drawn room and mood and leave two stars and a step on two other boards; draw the eleven floor plan symbols and state the $\frac{1}{4}$ inch = 1 foot rule; read a real floor plan and trace the	
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			traffic path; measure the classroom with a tape to the nearest half foot and draw it to scale; draw a bedroom to scale, place cut-outs in the four zones (sleep, work, store, move), and fix one path under 30 inches; start the floor-plan do-now stack (sketches 1 to 4) and the project folder; watch the four-minute CAD preview; take the vocabulary check	
4.3 Design a Room project	8 (days 106 to 113); the floor-plan do now runs from Lesson 4.10 through the last build day of 4.14 (days 104 to 112), and the stack is stapled and turned in with the project in 4.15	4.11 The Client Brief (1); 4.12 Plan: Scale Drawing and Traffic Flow (2); 4.13 Furnish It on a Budget (1); 4.14 Build the Model (3); 4.15 Pitch and Gallery Walk (1)	read the six client cards aloud and pick one and a path (Design Your Own Bedroom for a client, or Tiny House), alone or in a team of two; receive the project sheet and Rubric 04 on day 1; sort every card line into a need or a want and add two hidden needs; write the design problem	Project 04 - Design a Room.md, scored on Rubric 04 - Design a Room.md; presented and submitted in Lesson 4.15 (day 113)

			sentence and defend it to a pair with a different client; walk the 30-inch and 36-inch tape strips (with a borrowed walker if the nurse has one); draw the client's room outline at 1/2 inch = 1 foot from the card's numbers; draw the zones lightly and arrange 30 cut-out pieces, needs first; shade and measure every path in inches, circle the failures, draw the turning circle; peer review another client's plan on the checklist (mark, do not fix); revise, glue down, and write two changes and one rejected suggestion on the back; price every piece from the 40-item catalog in a new, secondhand, or	
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			DIY column; add sales tax and keep the running total at or under \$400 or \$1,500; fix the trade-off box and justify three decisions with the five-step model; measure and mark the shoebox or foam board to the plan; build furniture from scrap to scale; apply the color scheme and one texture; label GREEN and SAFE with a reason for this client and write the justification card; self-score criterion 4; fill the five-part pitch planner and rehearse once; pitch for 60 seconds to a classmate holding the client card and answer the one client question; walk the gallery and write four two-stars-and-a-step slips; fill the self-assessment row of Rubric 04	
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			with evidence; staple the do-now stack and turn in the stack in order	
4.4 Careers	2 (days 114 to 115)	4.16 Careers in Housing, Interior Design, and Sustainability (1); 4.17 Unit 4 Review and Test (1)	follow the Okafor room from idea to move-in as a chain of thirteen people; sort the thirteen career cards by path and again by workplace; compare three careers on the Project Chain sheet (what they do, a day in the life, the path, the pay range to be refreshed before teaching); rate ten “would I like this” statements and total them by career; choose two careers and write a fit rating with evidence from Project 04; hear the guest if one comes and write two questions research alone could not answer; name one local CTE	Assessment 04.3 - Unit 4 Test.md, given in the last 20 minutes of Lesson 4.17 (day 115); the self-check and the Project Chain sheet are daily work

			program; play Design a Room Grab (seven categories, 42 cards, 15 minutes); take the test	
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Day counts: $7 + 8 + 8 + 2 = 25$. The day-by-day placement is in Calendar 04.md.

The floor-plan do now, as the lessons set it. Every period from Lesson 4.10 Day 1 (day 104) through Lesson 4.14 Day 3 (day 112) opens with a timed floor plan sketch on quarter-inch graph paper from a prompt on the board: two minutes to draw, one to compare with a neighbor, and the sheet goes in the project folder. The lessons number the stack: Lesson 4.10 carries sketches 1 to 4, two a day (an opening do now and a closing sketch: this room from above, this room rearranged with a path shaded, a 10 by 11 bedroom at $1/4$ inch = 1 foot, any room in the building with a path shaded), Lesson 4.11 is sketch 5 (a 10 by 12 room at $1/4$ inch = 1 foot with a door and a window), 4.12 carries sketches 6 and 7 (a 12 by 14 room redrawn with the bed against a wall; a 6 by 9 entry with the path shaded and measured), 4.13 is sketch 8 (the client's room from memory), and 4.14 carries sketches 9, 10, and 11 (the client's room as a folded-down net; one wall as an elevation at $1/2$ inch = 1 foot; any room with a 5-foot turning circle). Eleven sketches. Each is scored complete, partial, or missing as daily work; the stack is stapled in order with the name on the first sheet and turned in with the project in Lesson 4.15, where the plan criterion of Rubric 04 can see the practice behind the final drawing. A team of two turns in one plan, one budget, one model, and two stacks. I saw a colleague run this for two weeks before a design project, and the final plans were the best I had seen from middle schoolers.

Flex lessons (drop first when time runs short): Lesson 4.9 (fold texture and pattern into the element list of Lesson 4.6 and move the mood board to the client brief day, Lesson 4.11, as the last ten minutes) and one build day of Lesson 4.14 (two build days; the labels and the justification finish in the first ten minutes of 4.15 and the pitches start at minute 12, which Lesson 4.14's own teacher notes already allow). For the 30-week version (Unit 4 at 21 days), also run Lesson 4.7 in one day (the eight-layout sort becomes homework) and combine Lessons 4.16 and 4.17 into one day (the project chain and the self-check in the first 18 minutes with the card sort as homework, the test in the last 20, the review game dropped). That is 25 minus 4, or 21. **Lessons that must not be cut:** 4.2 (the home safety hunt and the emergency plan). That is the safety lesson of this unit: a student who leaves without knowing where the smoke detector goes and two ways out of the bedroom is the student who gets hurt. Lesson 4.3 (repairs) and 4.10 (the floor plan) are the next two to protect, because the project depends on 4.10 and the unit's promise depends on 4.3; Lesson 4.3's own notes say never cut the station time, and Lesson 4.10's say never cut the measuring.

Semester version (10 days): the scope and sequence names safety and conservation, color, and a floor plan. Run 4.1 and 4.2 combined in one day (the seven jobs as the do now, the paper hunt and the emergency plan as the body; the family match goes home) (1); 4.3

in one day with three stations (screw board, plunger, breaker and bulb) and the tool ID as the do now (1); 4.4 in one day with the audit as homework (1); 4.5 with the quiz in the last 15 minutes (1); 4.6 and 4.7 combined (1); 4.8 in one day (the wheel and the demo in the first 15 minutes, the mixing and the pitch in the last 25) (1); 4.10 in one day with the vocabulary check in the last 12 minutes (1); the project as 4.11 folded into the first ten minutes of 4.12 Day 1 (the Okafor card for everyone, the problem sentence as the do now), then 4.12 as written, with the Budget Sheet from the catalog as homework and no model (2); 4.15 as the pitch and gallery walk with the unit test in the last 20 minutes (1). Total 10. The careers content moves onto the thirteen cards on the wall during the gallery walk.

Every-other-day version: the same 10 meetings, with the floor-plan do now assigned as homework on the off days (the prompt goes home on the grid sheet) and the audit done at home over a week.

7. Standards block

NYS module lines taught (every line, by code, with the lesson that teaches it)

- NYS EDM 1. Healthy, Safe, Sustainable Living Spaces a) Recognize that family, school, work, and community settings are all part of the individual's broader living space : Lessons 4.1, 4.11
- NYS EDM 1. Healthy, Safe, Sustainable Living Spaces b) Identify common safety hazards found in living spaces and describe ways to minimize risks to individuals, families, and household members : Lessons 4.2, 4.3, 4.17 (wording per the NYSED 2023 copy, see crosswalk E.5; 4.2 teaches the hazards half and 4.3 the maintenance half)
- NYS EDM 1. Healthy, Safe, Sustainable Living Spaces c) Formulate a safety plan for home, work, and community settings : Lessons 4.2, 4.3
- NYS EDM 1. Healthy, Safe, Sustainable Living Spaces d) Investigate ways to conserve natural resources in family, school, work, and community settings : Lessons 4.4, 4.5, 4.17
- NYS EDM 2. Environment and Interior Design a) Design shared environments and living space zones (for home, school, work, and/or community) which show respect for individual privacy needs and appreciation of large group social needs : Lessons 4.7, 4.8, 4.10, 4.11, 4.12, 4.14
- NYS EDM 2. Environment and Interior Design b) Create a floor plan designed to meet the needs of individuals, families, or communities : Lessons 4.10, 4.12, 4.14
- NYS EDM 2. Environment and Interior Design c) Incorporate the elements and principles of design into a design plan : Lessons 4.6, 4.7, 4.8, 4.9, 4.10, 4.12, 4.14, 4.15, 4.17
- NYS EDM 2. Environment and Interior Design d) Research how environmental and green home factors influence the design of functional, safe and aesthetic living spaces : Lessons 4.4, 4.5, 4.13, 4.14
- NYS EDM 2. Environment and Interior Design e) Predict future housing design incorporating technology and demographic data : Lessons 4.1, 4.16
- NYS EDM 3. Consumer Resources and Finance a) Construct a budget for implementation of a floor plan design : Lessons 4.13, 4.15, 4.17

- NYS EDM 3. Consumer Resources and Finance b) Use consumer skills to select household items : Lesson 4.13; Lesson 4.5 at unit level (the free, cheap, or money column of the Greener Room Swap)
- NYS EDM 4. Career Pathways a) Explain roles and functions of individuals engaged in environment, housing, and interior design careers : Lesson 4.16 (also checked on the test, Lesson 4.17)
- NYS EDM 4. Career Pathways b) Investigate education and training requirements and opportunities for career paths in environment, housing, and interior design fields : Lesson 4.16
- NYS EDM 4. Career Pathways c) Assess personal employability skills for careers in environment, housing, architecture, and interior design and evaluate personal suitability for these careers : Lesson 4.16

CTE theme lines used (text verbatim from the theme modules)

Lines a lesson page prints are listed with that lesson. Lines marked “unit level” come from the crosswalk’s EDM rows or are taken verbatim from the standards extract for content no lesson page prints; the unit carries them.

Sustainability:

- CTE SUS 1. Resources b) Explain how sustainability can be a factor in decision making : Lesson 4.1
- CTE SUS 1. Resources e) Investigate practices that promote stewardship of environmental resources : Lesson 4.4; Lesson 4.5 at unit level
- CTE SUS 1. Resources c) Define and give example of renewable and non-renewable resources : unit level (no lesson page prints it; Lesson 4.4 teaches conservation and cost, not the sources, and the line comes up only on review card C2 in Lesson 4.17; the state’s line reads as printed)
- CTE SUS 2. Environmental Management a) Explain the concept of “carbon footprint” : unit level (no lesson page prints it; the state’s own EDM sustainability activity, which Lesson 4.5’s connections note hands to the science teacher)
- CTE SUS 2. Environmental Management c) Describe ways individuals can adjust behaviors to minimize the impact of their choices on the environment : Lesson 4.5 (the four Rs and the Greener Room Swap)
- CTE SUS 2. Environmental Management f) Explain factors to consider when assessing the environmental impact of housing decisions : Lessons 4.5 (the five green factors), 4.14 (the GREEN label); Lessons 4.1, 4.13 at unit level
- CTE SUS 3. Energy Conservation a) List and describe energy sources used to power homes, school, workplaces, and communities : unit level (no lesson page prints it; see SUS 1. c) above)
- CTE SUS 3. Energy Conservation b) Categorize energy sources as renewable or nonrenewable : unit level (no lesson page prints it; see SUS 1. c) above)
- CTE SUS 3. Energy Conservation c) Understand and explain how individuals can limit the use of nonrenewable resources by reducing, reusing, and recycling : Lesson 4.5 at unit level (the four Rs; the page prints SUS 2. c) instead)

- CTE SUS 3. Energy Conservation d) Define “conservation” as it applies to energy : Lesson 4.4
- CTE SUS 3. Energy Conservation e) List and describe methods for energy conservation in the home, workplace, and community : Lessons 4.4, 4.17
- CTE SUS 3. Energy Conservation h) Practice energy conservation through a variety of classroom applications : Lesson 4.4 at unit level (the classroom audit with planted wastes; the page prints 3. d) and 3. e) instead)
- CTE SUS 4. Careers Related to Sustainability b) Analyze career paths within the natural resources, environmental management, and energy conservation fields : Lesson 4.16 (unit level; the energy auditor and sustainability consultant cards; the page prints CCO lines instead)

Problem Solving and Innovation:

- CTE PSI 2. Design Process (Proactive) a) Implement a formal design process to solve a given problem by : Lessons 4.6 to 4.15 and 4.17, with the sub-lines placed as the lesson pages print them:
 - CTE PSI 2. Design Process (Proactive) a) a. Defining the problem being addressed : Lessons 4.7, 4.11
 - CTE PSI 2. Design Process (Proactive) a) b. Defining criteria that must be met through the finished design : Lessons 4.8, 4.9, 4.11
 - CTE PSI 2. Design Process (Proactive) a) c. Defining constraints that must be adhered to : Lessons 4.8, 4.11
 - CTE PSI 2. Design Process (Proactive) a) d. Brainstorming and examining possible solutions : Lessons 4.7, 4.9, 4.10, 4.12
 - CTE PSI 2. Design Process (Proactive) a) e. Selecting the best solution for evaluation : Lessons 4.7, 4.12
 - CTE PSI 2. Design Process (Proactive) a) f. Developing and constructing a prototype or model of the selected design : Lesson 4.14
 - CTE PSI 2. Design Process (Proactive) a) g. Testing and evaluating the prototype and model against the design criteria and constraints : Lesson 4.7 (the gallery critique); Lessons 4.14, 4.15 at unit level (the Day 3 self-check and the client’s question)
 - CTE PSI 2. Design Process (Proactive) a) h. Optimizing the solution for best form and function : Lessons 4.6, 4.8, 4.10, 4.12, 4.17 (the test scenario)
 - CTE PSI 2. Design Process (Proactive) a) i. Evaluating their use of the design process and how it impacted their final solutions : Lesson 4.15 (unit level; the self-assessment row and the “step I got right” closure)
- CTE PSI 2. Design Process (Proactive) b) Demonstrate personal development of design skills through practice of these skills in a variety of classroom applications : Lessons 4.6, 4.9, 4.10; Lessons 4.7, 4.8 at unit level
- CTE PSI 3. Troubleshooting Process (Reactive) a) Implement a formal troubleshooting process to solve a given problem by a. Defining the problem being addressed; b. Identifying criteria and specifications for the desired outcomes or operation; c. Testing and evaluating to isolate the problem; d. Correcting the problem by implementing

changes or repairs; e. Validating that the corrective action produced desired outcomes or operation; f. Identifying strategies to prevent future problems : Lesson 4.3 (the six steps on the station cards and the maintenance calendar)

Health, Safety, and Wellness:

- CTE HSW 3. Personal Safety c) Summarize common causes of intentional and unintentional injury and describe associated prevention strategies : Lessons 4.2, 4.17
- CTE HSW 3. Personal Safety d) Explain how an orderly environment promotes reduction of accidents and injury : Lesson 4.1; Lessons 4.10, 4.12 at unit level
- CTE HSW 3. Personal Safety e) Develop safety plans and practice emergency responses : Lesson 4.2
- CTE HSW 4. Classroom and Workplace Safety c) Apply safety practices when using tools, machines, and equipment : Lesson 4.3; Lesson 4.14 at unit level (the glue gun station and the cutting station)
- CTE HSW 4. Classroom and Workplace Safety d) Wear appropriate protective clothing and gear when using tools, machines, and equipment : Lesson 4.3 (safety glasses at the screw board)
- CTE HSW 4. Classroom and Workplace Safety g) Understand and follow preplanned procedures in the case of accident or emergency : Lesson 4.2
- CTE HSW 5. Social-Emotional Learning a) Discuss how personal habits, environment, and heredity affect social-emotional health : Lesson 4.8 (color and mood)

Financial and Consumer Literacy:

- CTE FCL 4. Buying Goods and Services c) Develop a spending and savings plan (budget) based on income and expenses : Lessons 4.13, 4.17
- CTE FCL 4. Buying Goods and Services f) Apply a decision-making model to maximize consumer satisfaction when purchasing goods and services : Lesson 4.13; Lesson 4.5 at unit level
- CTE FCL 4. Buying Goods and Services g) Compare local, national, and global resources for obtaining goods and services : Lesson 4.13 (unit level; the three price columns)

Career and Community Opportunities:

- CTE CCO 2. Career Clusters a) Identify and use career resources to obtain information about careers and employment trends : Lesson 4.16
- CTE CCO 4. Career Plans b) Assess personal characteristics such as interests, abilities, and aptitudes in relation to characteristics associated with careers of interest : Lesson 4.16
- CTE CCO 1. The Work World c) Compare the similarities and differences among home, school, community, and work environments : Lesson 4.1 (unit level; the crosswalk row for EDM 1. a); the page prints HSW 3. d) and SUS 1. b) instead)
- CTE CCO 1. The Work World d) Identify changes likely to occur in home, school, and community environments that will likely change work environments in the future : Lesson 4.16 (printed on the lesson page; the crosswalk row for EDM 2. e))

- CTE CCO 3. Employability Skills a) Identify personal characteristics such as abilities, interests, and values and examine how they might impact career choices : Lesson 4.16 (unit level; the self-check)
- CTE CCO 4. Career Plans a) Examine potential career choices to determine knowledge, skills, and abilities associated with each : Lesson 4.16 (unit level; the Project Chain sheet)

Communication and Interpersonal Relationships:

- CTE CIR 1. Communication f) Demonstrate personal development of communication skills through practice of these skills in a variety of classroom applications : Lesson 4.15; Lesson 4.8 at unit level (the 30-second pitch)
- CTE CIR 2. Listening c) Show understanding of active listening and feedback techniques : Lesson 4.15 (unit level; the client's question and the feedback slips)

NYS FACS Learning Standards (Intermediate)

The state's EDM module lists Standards 2 and 3. Every lesson except 4.4, 4.5, 4.13, and 4.16 cites Standard 2 with PI B; 4.4 and 4.5 cite Standard 3 with PI B, 4.13 cites Standard 3 with PI C and 4.16 cites Standard 3 with PI E, as the crosswalk's EDM rows assign them.

Standard 2: A Safe and Healthy Environment. Students will acquire the knowledge and ability necessary to create and maintain a safe and healthy environment. Key Idea: Students will know the basic principles of home and community safety. They can demonstrate the skills necessary to maintain their homes and workplaces in a safe and comfortable condition. They can provide a safe and nurturing environment for themselves and others.

- PI B: Students know the basics of managing a safe and healthy home. (Lessons 4.1, 4.2, 4.3, 4.6 to 4.12, 4.14, 4.15, 4.17 as printed; the citation for EDM 1. a), 1. b), 1. c), and 2. a) through 2. e))

Standard 3: Resource Management. Students will understand and be able to manage their personal and community resources. Key Idea: Students will understand and be able to manage personal resources of talent, time, energy, and money and make effective decisions in order to balance their obligations to work, family, and self.

- PI A: Students understand how the family can provide for the economic, physical, and emotional needs of its members. (Lesson 4.1, unit level: housing needs across the lifespan and the three-generation family on the listings sheet)
- PI B: Students understand the resources available to them, make informed decisions about the use of those resources, and know some ways to expand resources. (Lessons 4.4, 4.5 as printed; the citation for EDM 1. d) and 3. b))
- PI C: Students are able to budget their time and money. (Lesson 4.13; the citation for EDM 3. a))
- PI E: Students identify their own abilities and interests as possible guides to career choice. (Lesson 4.16; the citation for EDM 4. a), b), c))

The EDM module PDF prints Standard 3 as “understand and can manage”; five other module PDFs and NYLearns print “understand and be able to manage,” and the crosswalk (E.2) says to use the latter. The PI letters are NYLearns codes; the NYSED print document lists the indicators as bullets. Several lesson pages carry their own note on this wording.

CDOS (Intermediate)

- CDOS 1 Career Development: Students will be knowledgeable about the world of work, explore career options, and relate personal skills, aptitudes, and abilities to future career decisions. (Lesson 4.16)
- CDOS 2 Integrated Learning: Students will demonstrate how academic knowledge and skills are applied in the workplace and other settings. (Unit level: Lessons 4.10, 4.12, 4.13, 4.17, where scale, area, and percent do the work an architect and a designer do every day; Lesson 4.4, watts and gallons)
- CDOS 3a Universal Foundation Skills: Students will demonstrate mastery of the foundation skills and competencies essential for success in the workplace. Sub skills used in this unit, as the lesson pages print them:
 - Basic Skills: Students listen to and read the ideas of others and analyze what they hear and read; acquire and use information from a variety of sources; and apply a combination of mathematical operations to solve problems in oral or written form. (Lessons 4.10, 4.12, 4.17; 4.4 at unit level)
 - Thinking Skills: Students evaluate facts, solve advanced problems, and make decisions by applying logic and reasoning skills. (Lessons 4.1, 4.6, 4.7, 4.8, 4.9, 4.11)
 - Personal Qualities: Students demonstrate an understanding of the relationship between individuals and society and interact with others in a positive manner. (Lesson 4.2)
 - Interpersonal Skills: Students demonstrate the ability to work with others, present facts that support arguments, listen to dissenting points of view, and reach a shared decision. (Lesson 4.15; 4.8 at unit level)
 - Technology: Students select and use appropriate technology to complete a task. (Lessons 4.3, 4.14; 4.4 (the plug-in meter) and 4.10 (the CAD preview) at unit level)
 - Managing Information: Students select and communicate information in an appropriate format (e.g., oral, written, graphic, pictorial, multimedia). (Unit level: Lessons 4.5, 4.9)
 - Managing Resources: Students understand the material, human, and financial resources needed to accomplish tasks and activities. (Lessons 4.4, 4.13)
 - Systems: Students understand the process of evaluating and modifying systems within an organization. (Lesson 4.5; Lessons 4.3, 4.4 at unit level; a home as a system)

National FCS 3.0 (LEAD FCS Education, 2018 to 2028; descriptors paraphrased from the 3.0 compilation, crosswalk A.5)

The EDM module block prints only the comprehensive standards 2.0, 3.0, and 11.0 and the content standards 11.1, 11.2, and 11.3. Every competency number below is confirmed in the 3.0 text (crosswalk B.4 and E.1) or in 02 Standards/STANDARDS SOURCES - web research.md, and the descriptor is paraphrased from the 3.0 text. Where the block's wording differs from 3.0, the 3.0 competency is the one paraphrased.

Competencies printed on lesson pages:

- 1.2.7 Examine what keeps school, work, and community settings safe and healthy. (Lessons 4.1, 4.2; the citation for EDM 1. a), b), c))
- 11.3.5 Examine how housing, interiors, and furnishings affect public health, safety, and welfare. (Lessons 4.2, 4.3; the citation for EDM 1. b))
- 11.6.5 Defend design choices in terms of the client's needs, including diversity and culture, and the design process. (Lessons 4.11, 4.15)
- 11.2.4 Design housing, interiors, and furnishings around how people behave, move, and measure. (Lessons 4.10, 4.11; the 36-inch path and the 60-inch turning circle)
- 11.4.3 Draw an interior space to scale using architectural symbols. (Lessons 4.10, 4.12; the citation for EDM 2. b))
- 11.4.4 Place furniture with design principles, traffic flow, activity, and the room's architecture in mind. (Lessons 4.7, 4.12, 4.14; the citation for EDM 2. a))
- 11.6.1 Work out what money and resources an interior improvement needs. (Lessons 4.13, 4.17; the citation for EDM 3. a))
- 2.1.5 Use consumer skills in choices about housing, utilities, and furnishings. (Lesson 4.13; the citation for EDM 3. b))
- 11.9.1 Show the knowledge and skill to recycle and redesign. (Lesson 4.14; 02 Standards/STANDARDS SOURCES - web research.md confirms only 11.9.1 under 11.9)
- 11.6.4 Critique a design plan against the client's needs, goals, and resources. (Lesson 4.15; the feedback slips)
- 11.1.1 Explain what people in housing and interior design careers do. (Lesson 4.16; the citation for EDM 4. a))
- 1.2.1 Examine possible careers to find what each one requires and offers in knowledge, skills, attitudes, and openings. (Lesson 4.16; the citation for EDM 4. c))
- 11.2.1 Judge how the elements and principles of design are used in homes and in commercial and residential interiors. (Lessons 4.6, 4.7, 4.8, 4.9, 4.17; the citation for EDM 2. c))
- 11.2.2 Examine how design elements and principles affect how a person feels in a space. (Lesson 4.8; the mood a color sets; Lesson 4.7 at unit level, the focal point)
- 11.2.3 Examine how design elements and principles shape how a space looks and how it works. (Lessons 4.6, 4.8, 4.9; from my Color in the Home citation)

- 11.3.1 Examine product information for floor and wall coverings, textiles, window treatments, furniture, lighting, and kitchen and bath fixtures and equipment. (Lesson 4.9; the swatch box)
- 11.4.2 Judge floor plans for efficiency and safety: zones, traffic patterns, storage, and the electrical, plumbing, ventilation, and heating systems. (Lesson 4.10; the crosswalk row for EDM 2. b))
- 2.2.3 Practice habits that conserve, reuse, and recycle so resources and the environment last. (Lessons 4.4, 4.5; the citation for EDM 1. d))
- 3.4.1 Look into where homes and businesses get energy and water, how waste is handled, and where pollution comes from. (Lesson 4.5; Lesson 4.4 at unit level; the citation for EDM 2. d))
- 3.4.3 Try out different ways to save energy and cut waste. (Lesson 4.4; Lesson 4.5 at unit level; the citation for EDM 1. d))

Competencies the unit carries at unit level (confirmed in the 3.0 text; from the crosswalk rows; no lesson page prints them):

- 3.4.5 Examine the parts government, culture, industry, and families each play in how energy gets used. (Lesson 4.4, the heat pump slide)
- 2.2.1 Examine the part individuals and families play in current environmental trends and issues. (Lesson 4.5)
- 2.2.2 Sum up the environmental trends and issues that affect families now and in the generations to come. (Lessons 4.1, 4.4)
- 11.4.5 Use building codes, universal design guidelines, and regulations when planning a space and placing furniture and furnishings. (Lessons 4.10, 4.12; the traffic flow rules)
- 11.4.6 Show graphic communication skills: sketching, slides, and CAD. (Lessons 4.10, 4.14; the daily sketch, the elevation, and the CAD preview)
- 11.3.3 Show the skills of measuring, estimating, ordering, buying, pricing, and repurposing. (Lessons 4.12, 4.13)
- 11.3.4 Weigh interior furnishings, finishes, fixtures, appliances, and equipment to give a client choices on cost and quality. (Lesson 4.13; the crosswalk row for EDM 3. b))
- 11.5.5 Predict where design in architecture, interiors, and furnishings is heading. (Lesson 4.16, printed on the lesson page; the crosswalk row for EDM 2. e))
- 11.1.2 Examine the career paths and the openings for a job or a business of your own. (Lesson 4.16; the contractor and the one-person design business)
- 11.1.3 Sum up the education, training, and credentials that housing and interior design careers require and the openings they offer. (Lesson 4.16; the three paths; the crosswalk row for EDM 4. b))
- 11.1.7 Examine the attitudes, traits, and values (responsibility, accountability, and doing the job well) that housing and interior design careers require. (Lesson 4.16; the crosswalk row for EDM 4. c))

Content standards cited at the content-standard level only, as the module block does:

- 11.2 Evaluate housing and design concepts and theories, including sustainability and universal design, in relation to available resources and options. (Lesson 4.1 prints it. The block prints “housing and design theories and concepts”; the 3.0 wording is cited.)
- 11.3 Apply residential and commercial interior design knowledge, skills and processes to meet specific design needs. (Unit level. The block prints “Apply interior design knowledge, skills, and processes”; the 3.0 wording is cited.)
- 5.2 Demonstrate planning, organizing, and maintaining an efficient operation of residential or commercial facilities. (Lesson 4.3 prints it; the Area 5 competencies are not in 02 Standards/STANDARDS SOURCES - web research.md, so 5.2 is cited as a content standard only.)

Not cited: 5.6.x. The 5.6 competencies were early candidates for this unit; in the 3.0 text 5.6 is laundering to industry standards, which belongs to Unit 5, and 02 Standards/STANDARDS SOURCES - web research.md holds no Area 5 competencies below the content-standard level. Lessons 4.2 and 4.3 say the same in their own notes. Nothing in this unit cites a 5.6 number. Credit LEAD FCS Education on the site.

Career Ready Practices (as the EDM module lists them; wording from the extract)

- CRP 2. Apply appropriate academic and technical skills (the EDM PDF prints “Apply appropriate and academic and technical skills”; the crosswalk (D.4) says to use the N&W wording)
- CRP 3. Attend to personal health and financial well-being
- CRP 5. Consider environmental, social, and economic impacts of decisions
- CRP 6. Demonstrate creativity and innovation
- CRP 7. Employ valid and reliable research strategies
- CRP 8. Utilize critical thinking to make sense of problems and persevere in solving them
- CRP 10. Plan education and career paths aligned to personal goals

Career Ready Practices are a secondary citation on the unit page only; the CCTC source text has not been fetched (crosswalk E.2).

8. Project

Name: Design a Room. Full sheet in Project 04 - Design a Room.md (handed out with Rubric 04 in Lesson 4.11 so nothing on the rubric is a surprise, reread in 4.14 Day 3 and 4.15); rubric in Rubric 04 - Design a Room.md.

The scenario: Every student is a designer with one client, and the client is not them. In Lesson 4.11 each table reads six client cards aloud; each card has a person, a room with its dimensions, a budget, must-haves, nice-to-haves, and one rule that cannot be broken. The lesson names the Okafor card (a grandfather who uses a walker, likes blue, needs a 36-inch path from the door to the bed and a chair with arms) and the Rivera card (a wheelchair user; every path is 36 inches and that number is not up for discussion), a shared sibling room (Luz, eight, is the sister who is not the main client), a baby’s room with a crib, a tiny house couple, and an entry with a coat hook that has to work for everyone. Two paths:

Design Your Own Bedroom for a client (a room refresh, \$400 cap: the walls and floor are there, the money buys furniture, storage, light, paint, and finish) or Tiny House (the state's own illustrative activity; the harder path, \$1,500 for the interior, with a kitchen and a bathroom to fit and a loft drawn as a second plan). A student may switch cards up to the end of Lesson 4.12 Day 1 with no penalty.

Deliverables (the turn-in stack, in the order Lesson 4.15 sets): (1) the client card; (2) the Needs and Wants List with every card line sorted, two hidden needs, and the design problem sentence "___ needs a room that ___ within ___"; (3) the scale plan at 1/2 inch = 1 foot (tiny house teams may use 1/4 inch on letter paper) with the door swing, windows, closet, scale, and client name, the 30 cut-out pieces glued down in zones, every path shaded and measured in inches, the turning circle if the client uses a wheelchair, the peer review checklist, and on the back two changes made and one suggestion rejected, with reasons; (4) the Budget Sheet: every piece on the plan priced from the 40-item catalog in a new, secondhand, or DIY column, sales tax at the county rate, a running total at or under the cap, the trade-off box, and three purchase justifications; (5) Rubric 04 with the self-assessment row filled, one score and one line of evidence per criterion; (6) the four feedback slips collected from the cup at the team's station; (7) each student's floor-plan do-now stack, stapled in order with the name on the first sheet. The model stays on the shelf: a shoebox or foam board box at a stated scale that matches the plan, furniture built from scrap to scale, the color scheme from Lesson 4.8's color card applied, one texture, a GREEN label and a SAFE label each with a reason for this client, and the justification card for the biggest design choice. The pitch is live: five parts in 60 seconds to a classmate holding the client card, then one client question answered in one sentence.

Steps by day: Day 1 (Lesson 4.11, day 106): the Okafor card read as a designer reads it; the six cards read aloud; the client, the path, and alone or a team of two chosen; the Needs and Wants List; the design problem sentence, swapped and questioned. Days 2 and 3 (4.12, days 107 and 108): the tape strips on the floor; the outline at project scale from the card's numbers; zones and cut-outs, needs first, traced or taped; the traffic flow rules; every path shaded and measured; the peer review checklist; revise, glue, and write the two changes and the rejection. Day 4 (4.13, day 109): the catalog, the three columns, the tax, the running total, the trade-off box, three justifications with the five-step model. Days 5 to 7 (4.14, days 110 to 112): the box measured and marked to the plan and the walls up; furniture from scrap in ninety seconds each, the floor covering, the color scheme and one texture; the GREEN and SAFE labels, the justification card, the self-score on criterion 4. Day 8 (4.15, day 113): the pitch planner and one rehearsal; two rounds of pitches at the stations with assigned clients; the gallery walk with four slips; the self-assessment row; the turn-in stack. Anything unfinished on the labels finishes in the first ten minutes of 4.15; late projects take the one-level drop the grading plan sets, never a zero.

Group or individual: alone or a team of two, chosen by the student with my veto (or assigned by client card in a class that cannot handle choosing). A team turns in one plan, one Budget Sheet, one model, one rubric, and two do-now stacks; every teammate says at least one part of the pitch. A student who cannot build submits the isometric grid drawing from the build guide in place of the model, scored on the same criterion.

Choice built in: the client card and the path; alone or a team; new, secondhand, or DIY for every piece; the model medium (a shoebox, foam board, or the isometric drawing); a larger scale (1 inch = 1 foot on 11 by 17) for a student who needs it; the pitch read from the planner, given to the teacher alone at the station, or given with a teammate reading while the student points; the color scheme from the student's own Lesson 4.8 card or one of the three on the build guide.

ELL and IEP supports: the client cards carry a picture line at the top (a walker, a desk, a coat hook, a crib, a tiny house, a wheelchair) so the client can be identified before the text is read; the Needs and Wants List, the plan guide, and the pitch planner carry sentence frames and the key words in Spanish (Vocabulary 04.md), to be verified by a native speaker; the problem sentence may be written in the home language first; the room outline can be pre-drawn and the furniture pre-cut; a printed tax table on the back of the Budget Sheet and a calculator; pre-cut wall pieces and pre-folded bed and desk blanks; a student who cannot use scissors or glue safely directs a partner's hands and owns the labels and the justification; two feedback slips instead of four; extended time into the next lesson at every step (the list into 4.12, the revision into 4.13, the budget into 4.14, the labels into 4.15, the self-assessment into 4.16).

How it is presented: Lesson 4.15. Stations are set so no two teams with the same client card sit together, and each team is assigned a "client," a student from another team who holds the card and listens as that person. Half the teams pitch while the other half are clients; at sixty seconds the timer sounds and the pitch stops; the client asks one question from the card ("Where do I put my walker at night?" "Can I reach the closet rod?"); the team answers in one sentence; then the halves swap. Then everyone walks: at four models that are not their own client, read the card, look at the plan and the model, and fill a slip with two stars (two things this design does for its client, specific) and one step. "I like it" does not count. A local interior designer, architect, or contractor as a guest client for the period, asking the questions, is the best version of this lesson (Section 12). The grade is the rubric: criterion 1 the client fit, from the Needs and Wants List and the pitch; 2 the plan; 3 the Budget Sheet; 4 the model and the design principles, from the model and the Day 3 clipboard check; 5 the pitch, from the clipboard checklist and the self-assessment, where a self-score within one level of mine on every line is one of the Exceeds conditions. A plain model with a plan that fits the client and a budget that adds up scores higher than a beautiful model that ignored the brief. I say that on day 1 of the project and again before the pitches.

9. Assessment plan

- **Formative (daily):** the do now (the floor-plan sketch from day 104 on), a check for understanding in every lesson (thumbs, a clipboard walk with a yes or no per student, a tool tray pass, a cold call), a 3-2-1 or one-line exit card most days, the station cards initialed at the repair stations (4.3), the audit checklist (4.4), the peer review checklist on the plan (4.12), the Budget Sheet exit card (4.13), and the Day 3 clipboard check on the model (4.14). All daily work is scored complete, partial, or missing.
- **Formal at the end of each topic:**

- Topic 4.1: Assessment 04.1 - Home Safety and Conservation Quiz.md, given in the last 15 minutes of Lesson 4.5 (day 97). It covers Lessons 4.1 to 4.5: the lessons say the seven jobs of a home, a scenario built from the two argued family matches, the breaker item, and the drain item are on it, with hazards by room, the emergency plan, the bulb and faucet math with numbers given, and the four Rs from the rest. Retake after a review for anyone under the file's cut score.
- Topic 4.2: two checks. The Color in the Home frosting lab (Lesson 4.8, day 102), scored on the Lab Rubric pattern printed inside that lesson (the three lesson objectives plus a safety and sanitation row, 1 to 4 each, 16 points; the pattern is Rubric 01 - Lab Rubric.md, and the descriptors come from my original: a labeled wheel, at least one visible tint and one visible shade on the mixing plate, a wall panel and a justification that match the Room Card's mood with a family or function reason). And Assessment 04.2 - Design Vocabulary Check.md, given in the last 12 minutes of Lesson 4.10 Day 2 (day 105): the elements, the principles, the color words, and the floor plan symbols, matched and applied to one room picture. The vocabulary check is the ticket to the client cards the next day.
- Topic 4.3: the project, scored on Rubric 04 - Design a Room.md after Lesson 4.15 (day 113): criterion 1 from the Needs and Wants List and the pitch, 2 from the plan, 3 from the Budget Sheet, 4 from the model and the Day 3 clipboard check, 5 from the pitch clipboard and the self-assessment row.
- Topic 4.4: the Project Chain sheet and the "would I like this" self-check, scored complete, partial, or missing as daily work, and the unit test.
- **Unit test:** Assessment 04.3 - Unit 4 Test.md, given in the last 20 minutes of Lesson 4.17 (day 115) after the 15-minute review game, Design a Room Grab (seven categories: elements, principles, safety, conservation, color, scale and plan, budget and careers). The test has three math items that allow a calculator (items 4, 11, and 14 are the energy, scale, and budget math), a scenario that asks for an optimized plan (PSI 2. a) h.), a careers item (EDM 4. a)), and a written response at the end; a reduced-choice version of the matching items exists for IEP plans, and extended time runs into the first ten minutes of Unit 5 Lesson 5.1. Retake once after a review.
- **Grade weights within the unit** follow the course grading plan: labs and projects 40 percent (the frosting lab on the Lab Rubric and the project on Rubric 04), topic assessments 25 percent (the quiz, the vocabulary check, the unit test), daily work 20 percent (including the eleven floor-plan sketches, the station cards, the maintenance calendar, the audit, the peer review checklist, the Project Chain sheet, and the self-check), participation and safety 15 percent (the repair stations, the frosting lab, and the glue gun and cutting stations on build days count here; a student who handles a tool the unsafe way after one reminder sits out that station and gets a zero for that day's participation and a conversation, the same rule as the kitchen).
- **Marking period note:** day 91 is the first day of marking period 3, so every check in this unit lands in MP3. Lesson 4.1's pre-assessment reads a Unit 4 pre-test given at the end of Unit 3 (three things a home must do, two kinds of housing, one tool, one small repair); that pre-test lands in MP2 and is not graded.
- **Missed days:** a student who misses the repair stations does the no-tools version (the steps ordered on cards from the station photos and the test named) and loses no

points; a student who misses the frosting lab does the paper palette (colored pencils on the wall panel sheet with the same justification), scored on the same rubric minus the sanitation row; a student who misses a do-now day sketches the missed prompt at home; a student who misses a project day picks up at the next step, with the list, the revision, the budget, or the labels finished in the extended-time window the lessons already allow. A student who misses the quiz, the vocabulary check, or the test takes it during the next do now.

10. Materials, budget, and room

Room assumptions: a regular classroom for most of the unit; the FACS room's counters and sink are used for the frosting lab (Lesson 4.8) and the drying shelf for the models, but nothing needs a stove. Tables that seat four (teams of three for the hunt, four to six at the repair stations, four for the card sorts); wall space for the gallery walk; a projector for the room photos, the floor plan symbols, and the CAD preview; one device per pair for the CAD preview and the grade 8 stretch on bls.gov, with paper for everything else. A flat shelf or cart, labeled by period, to store 14 to 28 shoebox models for three days. A glue gun station away from the traffic path with a heat-safe mat, and a teacher's cutting station for the craft knives (teacher use or supervised grade 8 use only).

Consumables and small tools, per section (as the lessons list them):

Item	Lesson	Amount	Cost	What to buy
Family cards and home listings	4.1	one handout per student, single-sided; scissors optional	school printing	Handout 04.01 - Match the Family to the Home.md; four large index cards labeled BABY, TEEN, GRANDPARENT, WHEELCHAIR USER for the corners
The described home and the emergency plan	4.2	one handout per student; highlighters, one per team of three	school printing; highlighters from the room	Handout 04.02 - Home Safety Hunt Floor Plan and Emergency Plan.md; the hunt is on paper, no staged hazards
Smoke detector and CO detector	4.2	one battery model of each, passed around	about \$15 and \$25 the first year, \$0 after,	hardware store

		and tested once; they go on the classroom wall after	or \$0 if the custodian lends them	
Repair tool tray	4.3	two Phillips and two flathead screwdrivers, pliers, an adjustable wrench, a 24-inch level, three tape measures, a sink plunger and a toilet plunger, a new furnace filter, an LED bulb and a small lamp, safety glasses (a class set of 8 to 10)	about \$60 the first year if bought, \$0 after;	hardware store
Station props	4.3	a scrap 2 by 6 board (about 18 inches) with pilot holes pre-drilled and a box of 1.25-inch Phillips wood screws plus a few flathead screws; a 5-gallon bucket with a sink strainer or a plastic tub with a drain hole; five taped lines and a book and a box to measure; a scrap board with a pivoting shelf bracket, a picture frame	about \$10 the first year (the screws are the only thing used up)	hardware store; the board offcut is free from any lumber yard

		on a string, and a nail; a large print of a breaker panel with one breaker in the middle position; a “60 W max” fixture label		
Station cards and the maintenance calendar	4.3	one handout per student	school printing	Handout 04.03 - Repair Station Cards and Maintenance Calendar.md
Plug-in electricity meter and two bulbs	4.4	one plug-in meter, optional, reused every year; one 60 W incandescent bulb and one 9 W LED to pass around	about \$20 the first year for the meter, or \$0 if the science department or the utility’s school kit lends one; about \$2 for the incandescent bulb, and the LED is on the Lesson 4.3 tray	the science teacher or an online order; a charger with nothing attached and a game console or cable box from the room to plug in
Energy audit handout, calculators, clipboards	4.4	one handout per student (the math page, the audit checklist, the twelve-fix table, the sample bill); calculators, one per pair; a clipboard per team for the Day 2 audit	school printing; calculators and clipboards from the room	Handout 04.04 - Home Energy Audit Checklist and Sample Bill.md; before Day 2, plant a few wastes in the room (a charger with nothing attached, lights on in an empty corner, a window open near a heater) and photograph

				the room for reset
The twelve sort items and four bins	4.5	twelve real items or pictures (a glass jar, a cardboard box, a banana peel, a plastic bag, a T-shirt with a hole, a working phone charger, a chipped mug, junk mail, a greasy pizza box, a wooden chair with a loose leg, a plastic water bottle, coffee grounds) and four bins or four labeled table corners: REDUCE, REUSE, RECYCLE, COMPOST; the handout one per student; the quiz one per student; calculators for the quiz	\$0 (props from the room and home; the peel and the grounds are props nobody handles)	Handout 04.05 - Greener Room Swap.md, Assessment 04.1 - Home Safety and Conservation Quiz.md
Elements Hunt sheet, the shoebox note, and the Focal Point Game kits	4.6, 4.7	4.6: the handout one per student, colored pencils in a shared cup per table, the shoebox note one half sheet per student; 4.7: the handout one	school printing; about \$3 in card stock for the furniture sets the first year, reused	Handout 04.06 - Elements Hunt.md, Handout 04.07 - Principles Sort and Focal Point Game Cards.md; the rooms are described on

		per student, five furniture card sets cut apart and in envelopes, five room outlines printed large (11 by 17 if you can), five focal point cards, glue sticks or tape, five markers		the slides and the handouts, so there are no photos to buy; the shoebox note goes home in 4.6
Frosting lab	4.8	one 16-ounce tub of white frosting per station for three stations of seven (each station splits into two sub-teams), one set of red, yellow, and blue gel colors per station (plus brown if the set has it), plus one tub and one gel set for the Day 1 demo, so nobody crosses the room with frosting on their hands; paper plates, eleven per station (a mixing plate and a wall panel per sub-team plus a practice swatch plate per student);	about \$32 to \$36 per class (Lesson 4.8: about \$9 to \$11 per station plus the demo kit)	grocery store; buy in the week it is used, never before a break; allergy check at the end of Lesson 4.7 and again against the roster before you buy

		plastic spoons, nine per station; parchment to cover the tables; wet wipes and paper towels; one trash bag per station; index cards for the do now and the 3-2-1		
Non-food substitute for the frosting lab	4.8	tempera paint in red, yellow, blue, white, and brown, paper plates, spoons or brushes	about \$12, lasts the year	for a student who cannot have the frosting near them, or for the whole class if the room is a no-food room; Lesson 4.8's no-kitchen alternative; clean-up moves from the trash bag to the sink
Swatch boxes and mood board supplies	4.9	one swatch box per table of four, about 40 pieces each (fabric scraps, wallpaper samples, tile and wood samples, paint chip cards, carpet squares if a flooring store donated any); card stock, one half sheet per student plus spares; scissors	\$0 to \$5 for the swatches; about \$2 in card stock	scraps from the sewing bin (Unit 5), free samples from a home store or the ReStore, old chip cards from a paint department

		one per student; glue sticks one per pair; scrap construction paper; colored pencils one set per table; the Room and Mood assignment cards cut from the handout		
Mood Board Planner and magazine pages	4.9, 4.14	one handout per student; old magazines and catalogs, about one per two students, for cutting	school printing; magazines free	Handout 04.09 - Mood Board Planner.md; ask students not to bring anything from home to cut up
Floor plan lesson kit	4.10	the five-part handout one per student, printed double sided; tape measures one per team of four (25-foot retracting is best); a manila or two-pocket folder per student, labeled, for the project folder that starts on Day 1; one real floor plan printed large with the address removed, and a half-size copy per pair;	school printing; folders about \$8 per section the first year if the school does not stock them; tape measures from the Lesson 4.3 tray and the custodian	Handout 04.10 - Floor Plan Symbols Graph Paper Guide and Furniture Cut-Outs.md, Assessment 04.2 - Design Vocabulary Check.md; a free browser floor plan tool open on the teacher device for the four-minute preview

		scissors one per pair and an envelope or paper clip per student for the cut-outs; the vocabulary check one per student for Day 2		
Quarter-inch graph paper and rulers	4.10 to 4.14	about 14 sheets per student (five in Lesson 4.10, seven for sketches 5 to 11, two for the project plan); 11 by 17 graph paper for tiny house teams if available; a class set of rulers marked in quarter inches; pencils with erasers	about \$10 for a ream of quarter-inch grid paper and a pad of 11 by 17; rulers from the room; an architect's scale ruler for the grade 8 stretch, one or two, about \$6	print the grid from the plan guide or buy a pad
Client cards, the Needs and Wants List, the project sheet, the rubric	4.11	six client cards on card stock, one set per table of six plus one extra set; the list, the project sheet, and Rubric 04, one each per student	school printing, about \$3 in card stock, reused	Handout 04.11 - Client Cards and Needs and Wants List.md, Project 04 - Design a Room.md, Rubric 04 - Design a Room.md
Plan guide, furniture cut-outs, traffic flow rules, peer review checklist	4.12	one four-page handout per student, the cut-out page (30 pieces at 1/2 inch = 1 foot) on card	school printing, about \$4 in card stock	Handout 04.12 - Scale Plan Guide Traffic Flow Rules and Peer Review Checklist.md;

		stock; scissors one per pair; glue sticks or removable tape; a 36-inch and a 30-inch strip of masking tape on the floor; a tape measure		
Furnishing catalog and Budget Sheet	4.13	the catalog (40 items, three prices each) printed once per pair on two sheets; the Budget Sheet one per student or team; calculators; a printed tax table on the back	school printing	Handout 04.13 - Furnishing Catalog and Budget Sheet.md; the prices are realistic 2026 Long Island figures to be refreshed before teaching;
Model materials	4.14	one shoebox per student or team, lid off (collected free from families and a shoe store; the note goes home in 4.6); foam board 20 by 30 inches, one sheet per two or three bases for students with no box; corrugated cardboard for extra walls; card stock, white and colored; scrap fabric, felt,	about \$2 to \$3.50 per model, so about \$45 per section for 14 to 16 team models and up to about \$90 if every student builds alone	craft store; ask a fabric store for remnants and a paint department for old chip cards

		paint chips, construction paper, magazine pages; glue sticks, white glue, low-temperature glue guns at one supervised station; masking tape; scissors; craft knives at the teacher's station only; rulers, pencils, markers		
Build guide and label cards	4.14	one build guide per team; two label cards (GREEN, SAFE) and a justification card per team; the isometric grid page for the no-materials alternative	school printing	Handout 04.14 - Model Build Guide and Label Cards.md
Pitch planner and feedback slips	4.15	one planner per team; four slips per student, cut; a cup per station; a clipboard sheet with the five pitch parts	school printing	Handout 04.15 - Pitch Planner and Feedback Slip.md

Career cards, the Project Chain sheet, the self-check	4.16	thirteen career cards, one set per table of four; the sheet and the self-check one per student	school printing, about \$4 in card stock, reused; the pay ranges are to be refreshed before teaching	Handout 04.16 - Housing and Design Careers Cards.md;
Review game cards and the test	4.17	42 question cards in seven categories, one set per team of four, plus the scoreboard; whiteboards and markers per team; the test one per student; calculators for the three math items	school printing, reused	Handout 04.17 - Unit 4 Review Game Cards.md, Assessment 04.3 - Unit 4 Test.md
Printing: the handouts for 4.4 to 4.10 (the audit, the Greener Room Swap, the Elements Hunt, the Principles Sort, the wheel and Room Cards, the Mood Board Planner, the five-part floor plan handout, the quiz, and the vocabulary check)	4.4 to 4.10	one copy of each per student	school printing	all from 05 Materials/Unit 04/
Sticky notes and painter's tape	4.6, 4.12, 4.15	one pad, one roll	already in the room	element labels on the walls, the floor strips, table numbers for the gallery walk; the

				Poison Control number stays on the board all unit
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Unit total for one section: about \$130 in consumables the first year if the custodian lends the tools and the detectors and the science department lends the plug-in meter, as the lessons suggest (frosting about \$34, model materials about \$45, graph paper, the scale ruler, and the project folders about \$24, card stock across five handouts about \$16, station screws and props about \$10, the incandescent bulb about \$2; the swatch boxes add up to \$5 if nothing is donated); up to about \$250 if the tool tray, the detectors, and the meter are all bought new. About \$80 after the first year (the frosting and the model materials are the two lines that repeat every section).

Equipment (already in a regular classroom; buy what is missing): projector and screen; a class set of rulers and a few calculators; a document camera or a phone on a stand for the toilet flapper photo and the wheel demo; a timer visible to the room; the reset cue; a drying shelf or cart for the models; a heat-safe mat for the glue gun station; a first aid kit within reach on repair day and build days (the screws, the screwdrivers, the craft knives, and the glue guns are the sharp and hot things in the unit; the safety rule is the same as the kitchen's: tool down before you talk, and a burn is treated as a burn, cool water and tell the teacher).

Handouts (in 05 Materials/Unit 04/, one per lesson, as the lesson pages name them):

- Topic 4.1: Handout 04.01 - Match the Family to the Home.md; Handout 04.02 - Home Safety Hunt Floor Plan and Emergency Plan.md; Handout 04.03 - Repair Station Cards and Maintenance Calendar.md; Handout 04.04 - Home Energy Audit Checklist and Sample Bill.md; Handout 04.05 - Greener Room Swap.md.
- Topic 4.2: Handout 04.06 - Elements Hunt.md; Handout 04.07 - Principles Sort and Focal Point Game Cards.md; Handout 04.08 - Color Wheel Room Cards and Pitch Sheet.md; Handout 04.09 - Mood Board Planner.md; Handout 04.10 - Floor Plan Symbols Graph Paper Guide and Furniture Cut-Outs.md.
- Topic 4.3: Handout 04.11 - Client Cards and Needs and Wants List.md; Handout 04.12 - Scale Plan Guide Traffic Flow Rules and Peer Review Checklist.md; Handout 04.13 - Furnishing Catalog and Budget Sheet.md; Handout 04.14 - Model Build Guide and Label Cards.md; Handout 04.15 - Pitch Planner and Feedback Slip.md.
- Topic 4.4: Handout 04.16 - Housing and Design Careers Cards.md; Handout 04.17 - Unit 4 Review Game Cards.md.

Slides: one slide outline per lesson, 4.1 through 4.17, in the same folder, Slides 04.MM - [Title].md (each lesson names its own with the lesson title; multi-day lessons mark Day 1, Day 2, Day 3 inside one outline). The site build renders every outline as an HTML deck.

No-kitchen, no-tools, and no-build plan for the whole unit: the frosting lab is the only food in the unit, and its no-kitchen alternative is tempera paint on the same paper plates with the same Room Cards, the same pitch, and the same rubric minus the sanitation row (the safety row still scores tool and paint handling). The repair stations have a no-tools version on the lesson page: five demonstration stations on the slides with a photo sequence for each, the steps ordered on cards and the test named; the screw and level stations can still run with a screwdriver, a level, and a scrap board from the custodian, which are safe in any room. The plunger station needs a bucket, not a sink. The energy audit runs on the FACS room itself if a student cannot do it at home. The model has the isometric grid drawing as its no-materials alternative, and the whole class can build on foam board if shoeboxes do not come in. The catalog is printed, so the budget day needs no device. Nothing in the unit needs a stove, a sewing machine, or the internet on the day.

11. Health and other-department coordination

Health: the health course covers fire safety, carbon monoxide, and poison safety in its own unit. Lesson 4.2 uses the same words (two ways out, a meeting spot, test the detector monthly, batteries yearly, replace at ten years, the Poison Control number 1-800-222-1222) and goes where health does not: fifteen hazards in a described home room by room, classified by type, and a plan that includes how a grandmother with a cane gets out and who carries the toddler. Agree on the detector rules and the emergency numbers so both rooms say the same thing. Anything a student says in 4.2 about an unsafe home (no detector, no heat, a broken lock, an adult who will not fix it) is handled as a referral to the counselor or the nurse, not a class discussion, and Lesson 4.1 already sets the rule that nobody is asked to describe where they live.

Science: the science teacher covers energy, heat transfer, and electricity in grade 6 or 8. Lesson 4.4 uses the same words (a watt is a rate, a kilowatt-hour is an amount, heat moves from warm to cold, insulation slows it) and a plug-in electricity meter the science department may own; Lesson 4.3's grade 8 stretch explains why a breaker trips (too much current for the wire) and what a GFCI does. FACS stays on the bill and the habit; the physics of it belongs to science.

Math: the math teacher covers ratio and scale drawings in grade 7 and area in grade 6. Lessons 4.10 and 4.12 use the same setup (a quarter inch or a half inch on the paper is one foot in the room; 36 inches is six squares at project scale) and the same words, so the skill transfers; the test day's do now is the two conversions side by side. The furniture cut-out sheet with real dimensions is a scale drawing problem set the math class can use, and the grade 8 stretch (floor area, the percent left open) is a percent problem. Sales tax at 8.625 percent in 4.13 is the percent-of-a-number skill from Unit 2.

Art: the art teacher teaches the color wheel and the elements and principles of design, often in grade 6. Lessons 4.6, 4.7, and 4.8 use the art room's vocabulary exactly (tint, shade, balance, emphasis or focal point, rhythm, proportion) so a student hears one set of words, and Lesson 4.14 borrows the isometric grid drawing. FACS adds the room: what the same principles do to a person who lives there. Share the Room Cards with the art teacher; the frosting lab can be a guest lesson in the art room if the FACS room is unavailable.

ELA: the design problem sentence (4.11) and the five-part pitch (4.15) are a claim with evidence (the path width, the total, the label); the ELA teacher can use the pitch planner as a persuasion outline. Share it so students see one structure.

Social studies: if the social studies course covers community, local government, or immigration, Lesson 4.1's housing types, zoning, and multigenerational households connect; the social studies teacher can show a zoning map of the town in exchange for the seven-jobs list. FACS stays on what a family needs from a space; why a town votes on zoning belongs to social studies.

Special education and the occupational therapist: Lesson 4.11 asks the school's occupational therapist, if there is one, for five minutes on why 36 inches and a 60-inch turning circle are the numbers for a wheelchair and what a walker user actually needs from a floor; the nurse's office may lend a walker or a wheelchair for the tape strips in 4.12.

Technology or engineering, if the school has it: the model is a prototype in the same design process the PSI theme describes; the two teachers can share the design process language.

The custodian or facilities manager: not a department, but the most important coordination in the unit. Lesson 4.3 asks the custodian to lend the tool tray, the plungers, the level, and the furnace filter, and for a five-minute visit on Day 1 to show the building's breaker panel door (not inside it) and the filter; Lesson 4.16 names the facilities manager as a possible ten-minute guest. That is the person in the school who lives the unit's big idea every day.

12. Connections (in-building and community)

- Inside the building: the art teacher, for Lessons 4.6 to 4.8 and 4.14. FACS gives the art room a set of Room Cards and a color lesson where the palette has a client and a function, a stack of mood boards for the art room wall, and a class that already knows the wheel when art needs it; the art teacher gives FACS the vocabulary students already know, the isometric grid lesson for the model alternative, a loan of tempera paint and brushes for the no-food version, and, if willing, ten minutes on the day of Lesson 4.6 to name the elements in one painting.
- Inside the building: the custodian or facilities manager, for Lessons 4.3 and 4.16. FACS gives the custodian a class that knows what a plunger and a breaker are for, a thank-you card from each student, and fewer flooded sinks and running faucets in the FACS room; the custodian gives FACS the tool tray on loan (screwdrivers, pliers, a wrench, a level, tape measures, both plungers, a furnace filter), the building's own numbers (what one running toilet costs, what the light bill is), a five-minute visit at the breaker panel door on 4.3 Day 1, and ten minutes as the guest in 4.16 if the schedule allows.
- Community: a Habitat for Humanity ReStore, for Lessons 4.5, 4.9, and 4.13. FACS gives the ReStore a class of future shoppers who know what it is and why it exists, a thank-you card from each student, and, if your school allows, a donation drive of one household item per family; the ReStore gives FACS a box of tile, wallpaper, fabric, and paint chip samples for the swatch box and the models, a photographed price list of ten

furnishings to keep the catalog's secondhand column honest each year, and a ten-minute talk (in person or on video) on what happens to a couch nobody wants.

- Community: a local architect, interior designer, or contractor, for Lesson 4.15 and 4.16. FACS gives the guest a gallery of client-driven room models to walk as the client, a class that has drawn eleven floor plans and measured every path, and a thank-you card from each student; the guest gives FACS one period as the client at the pitches asking the questions, a five-minute talk on what a day in the job is, two questions per student answered that research alone could not (the state's own illustrative activity), and one real floor plan (with the address removed) for Lesson 4.10 to read.
- Community: the utility company's energy education program, for Lesson 4.4. FACS gives the program a class that has done a home energy audit and computed the savings from one change, and a set of audit results with no names on them; the program gives FACS a classroom kit (LED bulbs, a watt meter on loan, a home energy checklist), a presenter for one period, and the current rebate numbers for heat pumps and insulation.
- Community: the local BOCES or your school's high school partner or feeder district, for Lesson 4.16. FACS gives the CTE program a class that has already drawn a plan to scale, priced a room, and pitched it, and one program name written on every self-check; the program gives FACS its course list for the trades on the career cards and, if willing, a student or an instructor for five minutes.

13. Vocabulary for translation

See Vocabulary 04.md: 69 Tier 3 terms across the four topics with a one-line student definition and a Spanish column, to be checked by a native speaker. The Topic 4.2 terms come from my Color in the Home vocabulary cards; the rest are new, including the words the lessons name (townhouse, manufactured home, clearance, turning circle, elevation, home inspector, energy auditor, property manager). The file also has sentence starters for the pitch, for the peer review, and for the repair stations in English and Spanish; the lessons name the frames they need (the family match, the calendar justification, the needs list, the pitch planner, the feedback slip), and the starters cover them.